

Assignment 4

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Student Name/Number: (doubleclick to edit)

Instructions

- Edit the above cell to include your name and student number.
- Submit just this `ipynb` to Canvas. Do not rename, it associates your student number with the submission automatically.

using Distributions, Plots, LaTeXStrings, LinearAlgebra, Statistics, Random, QuantEcon, NLsolve

Question 1

Take the example in [Simple Example with iid Income](#) where the stochastic process for income y_t follows

$$y_t = \mu + \sigma w_t \sim N(\mu, \sigma^2)$$

Under these assumptions, with quadratic utility and if $\beta R = 1$ the optimal policy is

$$\mathbb{E}_t\{c_{t+1}\} = c_t$$

Which we can use to derive for this income process if $F_0 = y_0 = 0$.

$$c_t = (1 - \beta) \left(\mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j y_{t+j} \right] + F_t \right) = \mu + (1 - \beta) \sigma \sum_{j=1}^t w_j$$

And finally

$$F_t = \sigma \sum_{j=1}^{t-1} w_j$$

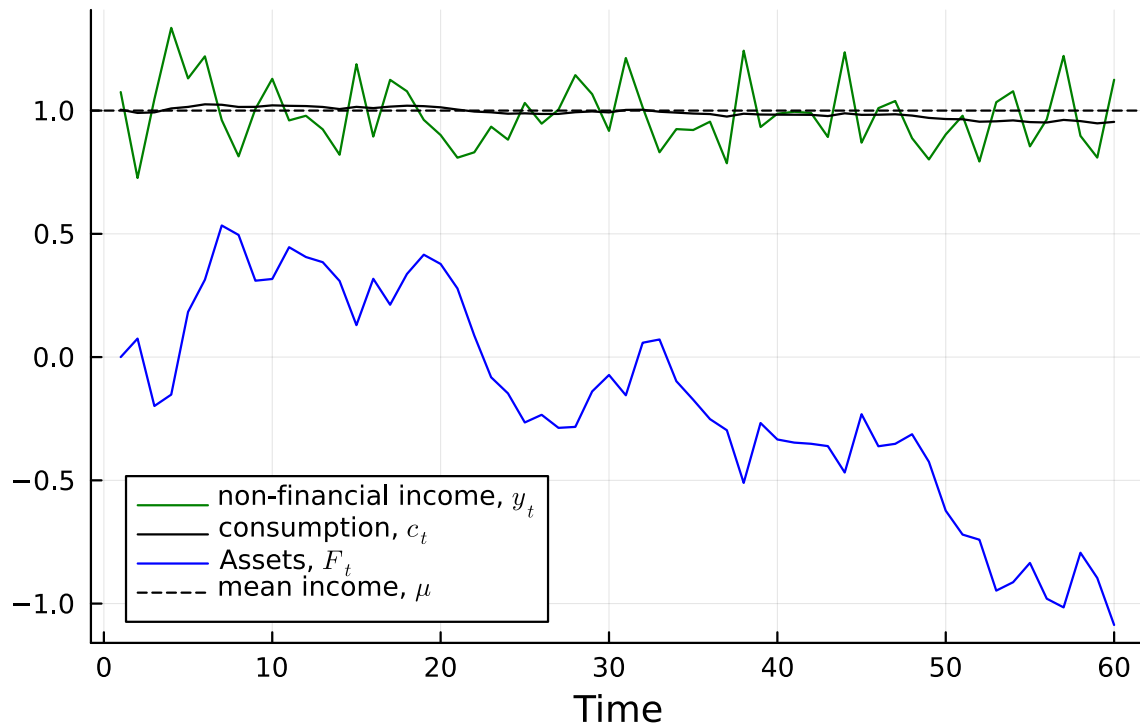
Part (a)

Code to simulate this consumption process for $t = 1, \dots, T$ is given below.

```
using Plots, Random
function simulate_iid_income(p, T; w = randn(T))
    w_sum = cumsum(w) # (w_1, w_1 + w_2, w_1 + w_2 + w_3, ... sum_{j=1}^T w_j)
    c = p.mu .+ (1 - p.beta) * p.sigma * w_sum # (c_1, c_2, ... c_T)
    y = p.mu .+ p.sigma * w # (y_1, y_2, ... y_T)
    F = [0.0; p.sigma * w_sum[1:end-1]] # (F_1, F_2, ... F_T)
    return (;w, F, c, y)
end

p = (;beta= 1.0 / (1.0 + 0.05), mu = 1.0, sigma = 0.15)
T = 60
res = simulate_iid_income(p, T)

plot(1:T, res.y, color = :green, label = L"non-financial income, $y_t$", xlabel="Time")
plot!(res.c, color = :black, label = L"consumption, $c_t$")
plot!(res.F, color = :blue, label = L"Assets, $F_t$")
hline!([p.mu], color = :black, linestyle = :dash, label = L"mean income, $\mu$")
```



Using this code, plot call the above where instead of randomly drawing each $w \sim N(0, 1)$ draw it from a distribution $w \sim N(0.5, 1)$ and plot the results

```
# edit your code here
```

Part (b)

Interpret the result of the previous graph with the standard one with well-specified results from the perspective of consumption smoothing and forecasting of income. In what ways is the behavior in the second case not optimal and why?

(double click to edit your answer)

Question 2

Continuing with income process defined in Part 1, rather than manually implementing this process, [use the LSS formulation](#)

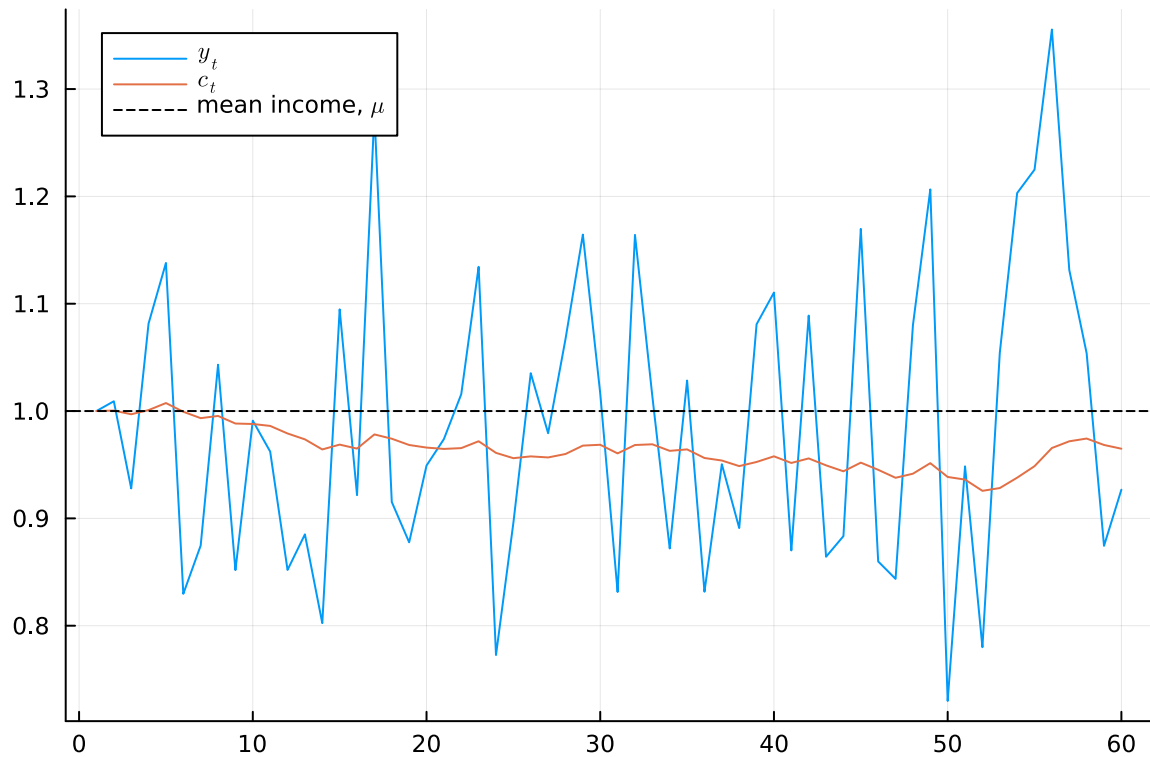
$$\begin{bmatrix} w_{t+1} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} w_t \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} w_{t+1}$$

$$y_t = \begin{bmatrix} \sigma & \mu \end{bmatrix} \begin{bmatrix} w_t \\ 1 \end{bmatrix}$$

From this, we can stack, etc. following [our notes](#)

The simulation from the code was

```
A = [0 0; 0 1]
G = [p.sigma p.mu]
C = [1; 0]
H = G*inv(I-p.beta*A)
A_tilde = [A zeros(2,1); H*(I-A) 1]
C_tilde = [C; 0]
G_tilde = [G 0; (1-p.beta)*H 1-p.beta]
x_tilde_0 = [0.0, 1, 0.0] #[w_0, 1, F_0]
lss_pi = LSS(A_tilde, C_tilde, G_tilde;
             mu_0 = x_tilde_0)
x, y = simulate(lss_pi, T)
plot(1:T, y[1,:];label=L"y_t", size=(600,400))
plot!(1:T, y[2,:], label=L"c_t")
hline!([p.mu], color=:black, linestyle=:dash,
       label = L"mean income, $\mu$")
```



Part (a)

Try calculating the stationary distribution by modifying the code below or trying with your existing code.

```
# modify here from previous results
# mu_x, mu_y, sigma_x, sigma_y = stationary_distributions(lss)
```

Explain why it does or does not converge below

(double click to edit your answer)

Part (b)

Consider a variation on the state space model above where instead of the state being above which puts the σ in the

$$C \equiv \begin{bmatrix} \sigma \\ 0 \end{bmatrix}$$

rather than in the G . Complete that state space model, and change the implementation below (copied from the above for simplicity)

```
# Modify here
A = [0 0; 0 1]
G = [p.sigma p.mu]
C = [1; 0] # replace with $C = [\sigma; 0]$ and fix the rest of the LSS/etc.
H = G*inv(I-p.beta*A)
A_tilde = [A zeros(2,1); H*(I-A) 1]
C_tilde = [C; 0]
G_tilde = [G 0; (1-p.beta)*H 1-p.beta]
x_tilde_0 = [0.0, 1, 0.0] #[w_0, 1, F_0]
lss_pi = LSS(A_tilde, C_tilde, G_tilde;
             mu_0 = x_tilde_0)
x, y = simulate(lss_pi, T)
plot(1:T, y[1,:];label=L"y_t", size=(600,400))
plot!(1:T, y[2,:], label=L"c_t")
hline!([p.mu], color=:black, linestyle=:dash,
       label = L"mean income, $\mu$")
```

